

WEEK 3 REPORT:

POWER BI STOCK MARKET ANALYSIS

Name: Monisha J

Roll No: E25DA029

Department: B.Sc Data Science and Artificial Intelligence

Introduction:

In Week 3, I used Power BI to analyze stock market data. I created a Date Table, Measures Table, DAX measures, relationships, and charts to understand stock prices and trading volume.

Dataset Used:

The dataset used for this project is SHOP_2015-05-21_2025-03-16.

The main columns are:

- Date
- Adj_close
- Open Price
- Close Price
- High Price
- Low Price
- Trading Volume

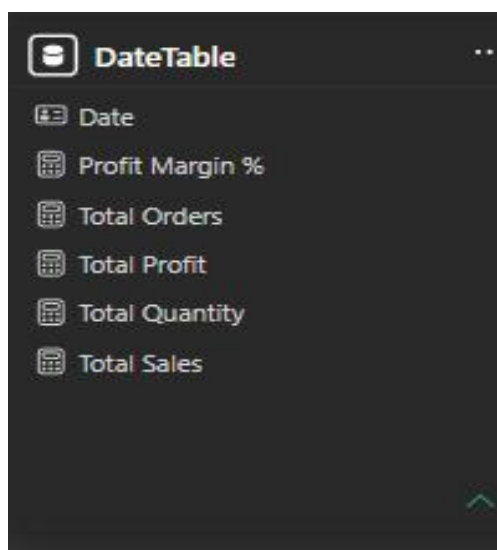
Creating the Date Table:

I created a separate Date Table using the CALENDAR() function.

The table contains:

- Date
- Day
- Month
- Month Number
- Quarter
- Year

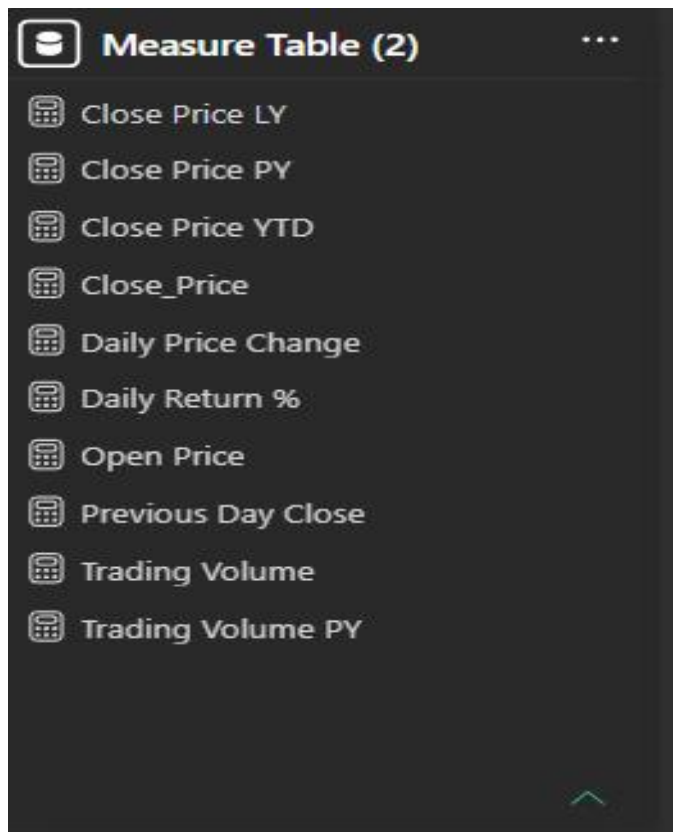
This table helps in analyzing the data based on dates.



Measures Table and Basic Price Measures

I created a separate Measures Table to store all the DAX measures, making them easy to find and manage. I also created measures such as Open Price, Close Price, Average

Open Price, Average Close Price, Highest Price, and Lowest Price to understand the stock prices.



Daily Price Analysis

I created measures to check the daily price changes.

The main measures are:

- Daily Price Change
- Daily Return %

These measures help to understand the change in stock price.

```
1 Daily Price Change =  
2 [Close_Price] - [Open Price]
```

```
Daily Return % =  
DIVIDE(  
    [Close_Price] - [Open Price],  
    [Open Price],  
    0)*100
```

Previous-Year Measures

I created previous-year measures using DAX.

The measures include:

- Close Price PY
- Trading Volume PY

These measures help compare the current data with the previous year's data.

```
1 Close Price PY =  
2 CALCULATE(  
3     [Close_Price],  
4     SAMEPERIODLASTYEAR('DateTable'[Date])  
5 )
```

```
1 Trading Volume PY =  
2 CALCULATE(  
3     [Trading Volume],  
4     SAMEPERIODLASTYEAR('DateTable'[Date])  
5 )
```

Creating the Table Relationship

I connected the DateTable with the SHOP table using the Date column. This relationship allows the stock data to be

analyzed based on different dates such as day, month, quarter, and year.

New relationship

From table

DateTable

Date	Day	Month	Month Number	Quarter	Year
21-05-2015 0...	21	May	5	Q2	2015
22-05-2015 0...	22	May	5	Q2	2015
23-05-2015 0...	23	May	5	Q2	2015

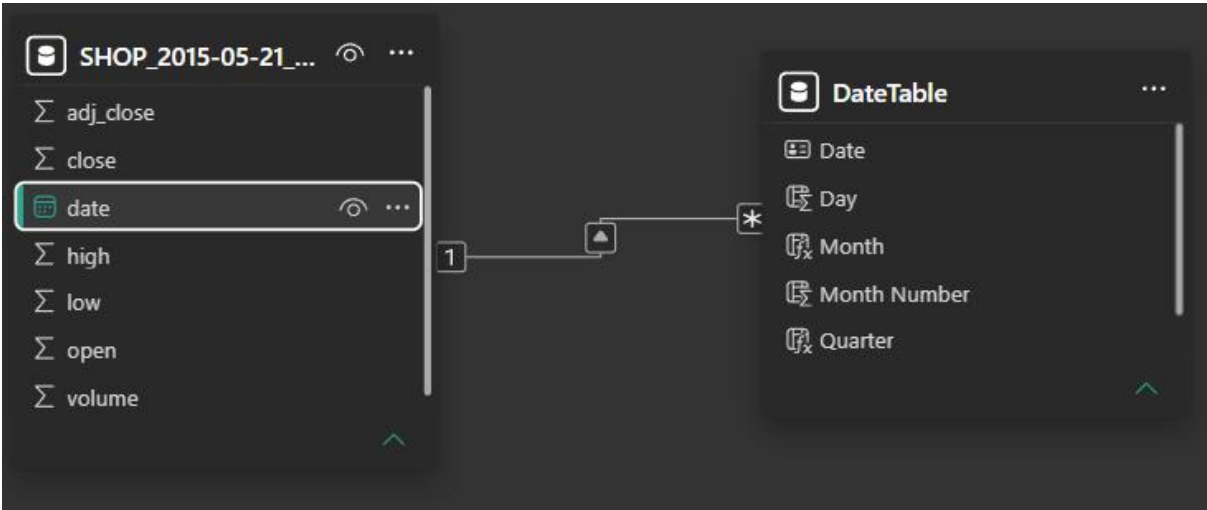
To table

SHOP_2015-05-21_2025-03-16

adj_close	close	date	high	low	open	volume
2.5680000782...	2.5680000782...	21-05-2015 0...	2.8740000724...	2.4110000133...	2.7999999523...	1230390
2.8310000896...	2.8310000896...	22-05-2015 0...	3.1099998950...	2.5999999046...	2.6070001125...	2841200
2.9649999141...	2.9649999141...	26-05-2015 0...	3.0339999198...	2.9079999923...	2.9800000190...	8202000

Save

Cancel



Final Dashboard

I created a final dashboard in Power BI to display the stock market data. It includes cards for Close Price, Low Price, Open Price, Trading Volume, and High Price. I also added charts for

Trading Volume by Year, Open and Close Price by Quarter, Trading Volume PY, and Close Price by Quarter. A month slicer was added to filter the dashboard.

